The sound we humans can hear with our ears is basically a modulation in air pressure. The airwaves push on our eardrums and the resultant movement is converted to electrical signals sent to our brain, which we sense as sound. Microphones sense sound in the same way (mechanical vibrations to electrical signals), and you may have noticed, it is the exact opposite of how a speaker works. Understanding the sensed sound is a higher level brain function. The mechanical analogue of recognizing sound would require a whole book on Fast Fourier Transforms. (For those interested, there is a great book titled "Who is Fourier?"). For now, it will suffice to imagine sound as vibration, and keep in mind that the information of any sound is contained in the patterns of the aforementioned vibrations. This chapter is dedicated to the science and technology behind audio information magically travelling and transforming from a file of bits and bytes on a server halfway across the earth to comprehensible airwaves in your inner canal.

Transmission Mediums

Sound, being nothing but oscillations, needs a medium in order to traverse across space. Just before we hear anything, with the exception of bone conduction, the medium is always air. That is why there is the eerie silence of outer space, which has no medium. Most broadly, transmission media can be categorized into two types, pertaining to the usability of the technology - Wired and Wireless.

It's a wireless world!

Wired media needs very little explanation as it is just a conducting material that transmits electrical signals from one end to another. Though it is a physical media, it should be noted that the actual sound vibrations are not being transmitted directly (as is the case for a cup-string-cup telephone). The sound waves are encoded into electrical signals, the kind that (usually) a speaker can understand. Then again, there is the whole analog vs digital issue, which will be covered in a subsequent chapter. The audio cables consist of two or more individual wires, one for the